

NICKOLAS J. VELA

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EDUCATION

UNIVERSITY OF CALIFORNIA, SANTA CRUZ

Santa Cruz, CA

B.S. Computer Science & B.A. Business Management Economics

Expected Jun 2028

Cumulative GPA: 3.86/4.00

Relevant Coursework: Data Structures & Algorithms, Artificial Intelligence, Machine Learning, C Programming, Python, Computer Architecture, Computer Systems, Linear Algebra, Multivariable Calculus

EXPERIENCE

TECH4GOOD LAB

Remote

Web Developer

June 2026 – Present

- Selected for a competitive, industry-aligned software development program that trains students to contribute to real undergraduate research at the Tech4Good Lab.
- Building responsive, component-based features for a full-stack web app using Angular, TypeScript, HTML, and CSS.
- Collaborating with a partner team through pull requests, peer code reviews, and test-driven development, mirroring professional engineering workflows.

OVERTURE MAPS FOUNDATION

Hybrid

Software Engineering Intern

Jan 2026 – March 2026

- Led a 4-person research team, directing project strategy, coordinating individual research approaches, and organizing the team's final presentation to sponsor company's lead engineers.
- Built an AI pipeline combining YOLOv8 street-view detection with a ResNet-18 satellite CNN fallback to predict building entrance locations, increasing entrance coverage from 5.8% to 56% across a 500-point national ground-truth dataset.
- Presented research findings to engineers and leadership from Overture Maps Foundation sponsor companies including Meta, Amazon, Microsoft, Niantic, and TomTom.

UNIVERSITY OF CALIFORNIA, SANTA CRUZ

Santa Cruz, CA

Resident Assistant

Aug 2025 – Present

- Served as a community leader and contact point for 50+ residents, creating an inclusive and supportive living environment.
- Developed strong leadership and communication skills through weekly meetings, one-on-one mentoring, and facilitating group discussions.

PROJECTS

BOARED | *TypeScript, React Native, Expo SDK 54, Supabase, PostgreSQL, Stripe Connect*

Feb 2026 – Present

Co-Founder

- Co-founded a social media app that lets college students discover local parties, shows, and events all in one place.
- Designed and shipped end-to-end a full-stack, cross-platform mobile app in TypeScript using React Native and Expo, owning frontend and database architecture.
- Building and evolving a PostgreSQL database with row-level security to support events, ticketing, RSVPs, messaging, follows, and content moderation.
- Built a full messaging system supporting DM, group, and event conversation types with RLS-enforced access control.

PORTFOLIO WEBSITE | *JavaScript, CSS, HTML*

Oct 2025

- Engineered a fully responsive and mobile-friendly website using HTML, CSS, and JavaScript, ensuring seamless performance across devices and screen sizes.
- Optimized user engagement through interactive elements including draggable physics objects, 3D CSS transforms with perspective effects, and debounced scroll event handlers for improved performance

ADDITIONAL

Languages: Python, C, C++, Swift, TypeScript, JavaScript, SQL, HTML, CSS, Shell Script

Technologies: React Native, Expo, Supabase, PostgreSQL, SwiftUI, Bash, Git, Vim, Firebase, VS Code, Xcode, Claude Code, Stripe Connect

Libraries/Frameworks: Pandas, NumPy, Scikit-Learn, Matplotlib, PyTorch, Ultralytics